

DATA ENGINEERING + ML PORTFOLIO PROJECT

Goodwill Donation Analytics & Revenue Forecasting Platform

Analyzing Form 990 financials and macroeconomic variables

Leslie Franchs

Strategic Systems Engineering Portfolio

Background & Practical Grounding

1.5 Years On-the-Ground Context

- 👉 **Firsthand Retail Observations:** Spent over 18 months observing daily donation intake, logistics, sorting operations, and community program execution.
- 📈 **Connecting Practice to Data:** Built this platform to link grassroots nonprofit exposure to enterprise data engineering and statistical modeling.
- 💡 **Core Business Question:** Can macroscopic economic indicators and regional factors effectively forecast thrift-store revenue performance?



📷 Local File: IMG_3014.JPG — On-the-ground volunteer service at local Goodwill operations.

Goodwill volunteer service — firsthand exposure to donation intake and local operations.

Project Executive Summary

🎯 Objective & Methodology

The goal was to forecast regional Goodwill revenues using historical Form 990 nonprofit filings integrated with localized economic indicators.

We developed an automated Python ETL process coupled with a PyTorch linear regression training pipeline to compare raw baseline models against strategically refined feature selections.

📊 Key Results & Insights

Refined Projection: The final model (with data-leakage features removed) forecasts steady future target revenue of ~\$70.5M.

Limitation & Stability: Initial model iterations used 14 years of direct historical data (2011–2024). To counter instability, we engineered **3,000 synthetic observations** to stabilize model training.

Primary Financial & Economic Inputs



Goodwill Financials

IRS Form 990 filings extracted from ProPublica Nonprofit Explorer. Target variables include annual Revenue, Operating Expenses, Net Income, and total Net Assets.



FRED Economic Data

Federal Reserve Economic Data (FRED) tracking annual US Unemployment rates and Consumer Price Index (CPI) values, aligned directly to financial reporting periods.



Global Oil Benchmarks

FRED Brent Crude oil annual average prices. Included specifically to test the hypothesis of whether fuel and heating costs influence counter-cyclical thrift retail behavior.

System Architecture & Pipeline



Historical Base Dataset (2011–2024)

Year	Revenue (\$)	Expenses (\$)	Unemp (%)	CPI	Oil (\$/bbl)	Net Assets (\$)
2024	92,588,656	79,403,516	4.0	318.99	80.52	71,543,655
2023	70,964,985	67,699,789	3.6	308.38	82.49	57,810,203
2022	61,913,793	63,836,428	3.7	294.31	100.93	54,284,446
2021	66,436,715	55,346,277	5.4	277.25	70.86	57,635,094
2020 (Covid)	77,145,526	52,306,762	8.1	267.71	41.96	46,385,330
2019	60,908,886	60,671,880	3.7	263.21	64.30	23,453,949
2018	58,110,226	59,731,817	3.9	257.56	71.34	22,613,609
2017	58,045,952	54,824,634	4.4	252.15	54.12	24,739,907

**Note: Final model targets Revenue while strictly omitting Net Income from features to guarantee statistical validity.*

Model Evolution & Leakage Mitigation

⚠ Version 1: Baseline First Run

Included **Net Income** as a model feature predictor.

The Issue: Produced an artificially perfect R^2 of 1.000.

Leakage Vector: Net Income is mathematically derived directly from Revenue and Expenses:

$$\text{Net Income} = \text{Revenue} - \text{Expenses}$$

✅ Version 2: Refined Robust Model

Net Income was completely removed from the training features.

Feature Inputs Used: Total Expenses, CPI, Unemployment, Brent Oil Prices, and Net Assets.

Validation: Demonstrates disciplined data engineering, active diagnostic review, and robust real-world forecasting.

Synthetic Data Expansion Strategy

3,000

Generated Synthetic Observations

Addressing Sample Constraints

The Challenge: Model training over only 14 historical data points from 2011–2024 left the PyTorch regression network highly prone to overfitting and unstable projection outputs.

The Solution: We simulated 3,000 synthetic data observations strictly anchored inside authentic historical statistical variances and multi-feature correlations.

Cautionary Interpretation: These 3,000 expanded rows behave as simulated validation data for structural training stability rather than genuine independent history.

Final Model Performance & Output

Forecasting Revenue

Utilizing our optimized PyTorch linear regression network, the system generates clean predictive results under future target parameters:

PREDICTED FUTURE ANNUAL REVENUE

\$70.5M

This prediction falls squarely within realistic, defensible historical limits unlike first-pass models which yielded extreme negative outliers.

Model Feature Coefficients



Model Coefficients & Business Logic

Variable	Coefficient	Expected?	Business Interpretation	Comments
Expenses	0.9475	✔ Yes	For every \$1 increase in Expenses, predicted Revenue increases by approximately \$0.95.	Strongly expected. Larger operations generally generate more revenue.
CPI	130,000	⚠ Partially	For every 1-point increase in CPI, predicted Revenue increases by approximately \$130,000.	Expected to be positive, but may also be capturing inflation and the passage of time.
Unemployment	3,513,000	📘 Initially No → ✔ After Review	For every 1 percentage-point increase in Unemployment, predicted Revenue increases by approximately \$3.51M.	Economically plausible for thrift retailers as budget-seeking grows.
Brent Oil	-128,800	✔ Yes	For every \$1 increase in Brent Crude Oil price, predicted Revenue decreases by approximately \$128,800.	Higher fuel costs reduce consumer disposable retail spending.
Net Assets	0.1987	✔ Yes	For every \$1 increase in Net Assets, predicted Revenue increases by approximately \$0.20.	Corresponds to more extensive store networks and logistics.

Slide Conclusion: Four of the five coefficients behaved largely as expected. The unemployment coefficient was initially surprising but became economically plausible when considered within the context of Goodwill's thrift retail model.

Business & Thrift Interpretation

Counter-Cyclical Thrift Behavior

Positive Unemployment Correlation: Traditionally, a rising unemployment coefficient is concerning. However, for thrift retail, this can reflect a logical business reality.

Trading Down: Under macroeconomic pressure, middle-market consumer groups often shift behavior away from full-price department stores down to lower-cost Goodwill options.

Firsthand volunteer observations of elevated storefront crowds during tighter economic cycles support this modeling hypothesis.



The "Jeans" Metaphor

"A decent pair of jeans at Goodwill costs a small fraction of the retail price at traditional brand storefronts, prompting organic consumer migration."

Core Lessons & Strategic Takeaways

-  **Data Quality Trumps Complexity:** Initial models produced invalid negative revenue forecasts when future inputs sat outside historical ranges. Rigid bounds and synthetic data validation proved essential.
-  **Identifying Hidden Leakage:** Net Income had to be structured out of features. High model accuracy ($R^2 = 1.0$) is almost always a flag for data leakage rather than superb design.
-  **Logic Over Metrics:** Selecting and fine-tuning variables based on physical operational business insight yields far more robust and reliable predictive platforms than raw algorithms.

THE NEXT PHASE

Future Development

The full codebase, datasets, and pipeline scripts are fully documented and public. Moving forward, the goal is to integrate real-time point-of-sale operational data from state-level databases into interactive Power BI dashboards.

PORTFOLIO LINK

 github.com/lesliefranchs2017

COLLABORATE

 leslie_franchs@yahoo.com